



AeroFuel Manager

Aircraft Fuel Cost & Trip Management System

■ PROVISIONAL AUTOMATED TEST REPORT ■

Report Generated: 17 April 2026, 11:49:01 PM

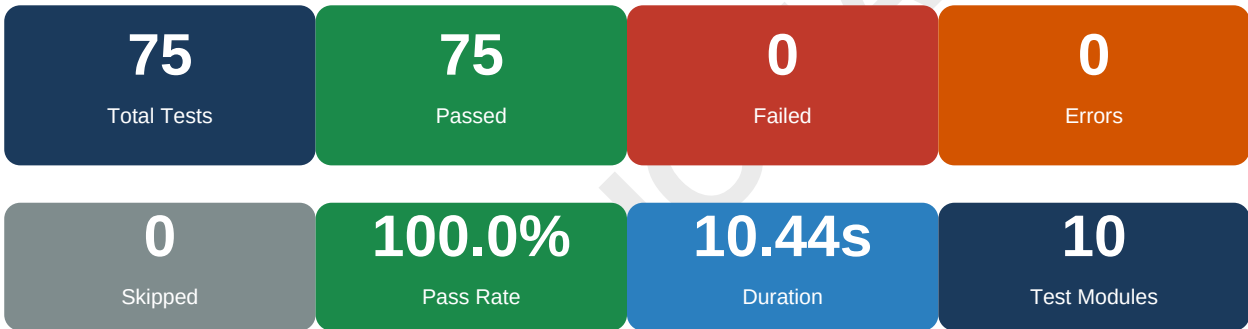
Test Environment: In-Memory SQLite (Isolated)

Framework: Python unittest + Flask Test Client

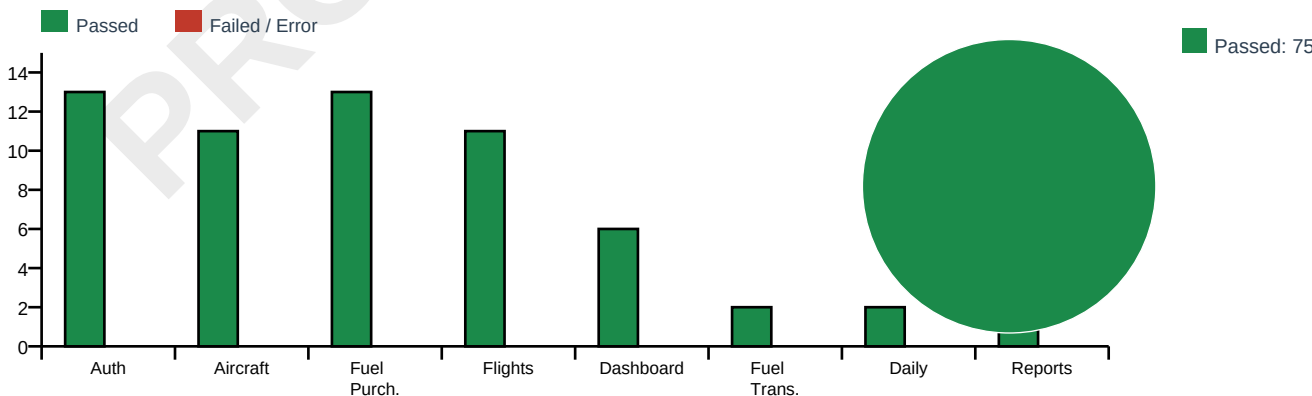
Backend Version: Flask 3.0.3 / SQLAlchemy 2.0

Report Status: ■ PROVISIONAL — Not for deployment sign-off

Executive Summary



Visual Overview



Left: Pass vs Fail per test module | Right: Overall result distribution

Module Summary

Module	Total	Passed	Failed	Errors	Skipped	Pass Rate	Status
■ Authentication	13	13	0	0	0	100%	PASS

▼ ■ Aircraft Management	11	11	0	0	0	100%	PASS
■ Fuel Purchases & Pricing	13	13	0	0	0	100%	PASS
■ Flight Trip Management	11	11	0	0	0	100%	PASS
■ Dashboard & KPIs	6	6	0	0	0	100%	PASS
■ Fuel Transactions	2	2	0	0	0	100%	PASS
■ Daily Usage	2	2	0	0	0	100%	PASS
■ Reports	2	2	0	0	0	100%	PASS
■ ■ Business Logic & Models	8	8	0	0	0	100%	PASS
■ Security & Access Control	7	7	0	0	0	100%	PASS
TOTAL	75	75	0	0	0	100.0%	PASS

PROVISIONAL

Detailed Test Results

■ Authentication

13/13 passed

#	Test Case	Status	Duration (s)
1	Create User As Admin	PASS	0.1161
2	Create User Duplicate Username	PASS	0.0021
3	Create User Invalid Role	PASS	0.0023
4	Get Profile	PASS	0.0026
5	Health Check	PASS	0.0004
6	List Users As Admin	PASS	0.0022
7	List Users As Operator Forbidden	PASS	0.0005
8	Login Missing Fields	PASS	0.0003
9	Login Success Admin	PASS	0.1172
10	Login Success Operator	PASS	0.1190
11	Login Unknown User	PASS	0.0025
12	Login Wrong Password	PASS	0.1214
13	Profile Without Token	PASS	0.0007

✈ ■ Aircraft Management

11/11 passed

#	Test Case	Status	Duration (s)
14	Aircraft Stats	PASS	0.0050
15	Create Aircraft As Admin	PASS	0.0107
16	Create Aircraft As Operator Forbidden	PASS	0.0007
17	Create Aircraft Duplicate Id	PASS	0.0023
18	Create Aircraft Missing Required	PASS	0.0006
19	Delete Aircraft Soft	PASS	0.0134
20	Get Aircraft By Id	PASS	0.0019
21	Get Aircraft Not Found	PASS	0.0017
22	List Active Only	PASS	0.0027
23	List Aircraft	PASS	0.0017
24	Update Aircraft	PASS	0.0053

■ Fuel Purchases & Pricing

13/13 passed

#	Test Case	Status	Duration (s)
25	Average Price	PASS	0.0046
26	Create Purchase Invalid Fuel Type	PASS	0.0007
27	Create Purchase Jet	PASS	0.0431

28	Create Purchase Missing Field	PASS	0.0024
29	Create Supplier	PASS	0.0154
30	Create Supplier Missing Name	PASS	0.0014
31	Current Price	PASS	0.0086
32	Filter By Fuel Type	PASS	0.0109
33	Get Single Purchase	PASS	0.0047
34	List Purchases	PASS	0.0085
35	List Suppliers	PASS	0.0035
36	Predict Price	PASS	1.2323
37	Price Trend	PASS	0.0029

Flight Trip Management

11/11 passed

#	Test Case	Status	Duration (s)
38	Cancel Completed Flight	PASS	0.0033
39	Cancel Flight	PASS	0.0294
40	Complete Flight	PASS	0.0288
41	Create Flight Aircraft Not Found	PASS	0.0028
42	Create Flight Auto Fuel	PASS	0.0123
43	Create Flight Duplicate Number	PASS	0.0023
44	Create Flight Missing Fields	PASS	0.0007
45	Filter By Status	PASS	0.0072
46	Get Flight Not Found	PASS	0.0021
47	Get Single Flight	PASS	0.0027
48	List Flights	PASS	0.0064

Dashboard & KPIs

6/6 passed

#	Test Case	Status	Duration (s)
49	Aircraft Consumption	PASS	0.0052
50	Fuel Consumption Chart	PASS	0.0029
51	Kpis	PASS	0.0113
52	Kpis Fuel Stock Present	PASS	0.0063
53	Kpis Unauthenticated	PASS	0.0005
54	Price Chart	PASS	0.0023

Fuel Transactions

2/2 passed

#	Test Case	Status	Duration (s)
55	List Transactions	PASS	0.0078

56	Transactions Unauthenticated	PASS	0.0005
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■ Daily Usage

2/2 passed

#	Test Case	Status	Duration (s)
57	Daily Usage Unauthenticated	PASS	0.0006
58	List Daily Usage	PASS	0.0030

■ Reports

2/2 passed

#	Test Case	Status	Duration (s)
59	Reports Overview	PASS	0.0006
60	Reports Unauthenticated	PASS	0.0004

■ Business Logic & Models

8/8 passed

#	Test Case	Status	Duration (s)
61	Aircraft To Dict	PASS	0.0020
62	Daily Fuel Usage To Dict	PASS	0.0022
63	Flight Fuel Auto Calculation	PASS	0.0040
64	Flight Insufficient Inventory	PASS	0.0069
65	Fuel Purchase Total Cost	PASS	0.0019
66	Inventory Update On Purchase	PASS	0.0430
67	User Password Hashing	PASS	0.3549
68	User To Dict	PASS	0.0025

■ Security & Access Control

7/7 passed

#	Test Case	Status	Duration (s)
69	Invalid Token	PASS	0.0008
70	No Token Aircraft	PASS	0.0004
71	No Token Flights	PASS	0.0005
72	No Token Fuel Purchases	PASS	0.0004
73	Operator Cannot Create Aircraft	PASS	0.0007
74	Operator Cannot Delete Aircraft	PASS	0.0005
75	Operator Cannot List Users	PASS	0.0005

■ PROVISIONAL REPORT — IMPORTANT NOTICE ■

This document is a **PROVISIONAL AUTOMATED TEST REPORT** generated by the AeroFuel Manager Automated QA Framework. It is intended for **internal development review purposes only**.

Scope of Testing:

This report covers automated unit and integration tests for the backend REST API of the Aircraft Fuel Cost & Trip Management System. Tests are executed against an isolated **in-memory SQLite database** using seeded fixture data. Results may not reflect the behavior of the system against a production PostgreSQL database.

Limitations:

- Frontend (React) components are **not covered** by this report
- End-to-end browser tests are **not included**
- Load/stress/performance tests are **not conducted**
- Database migration tests are **not conducted**
- Tests marked ERROR may indicate environment issues, not application defects

Intended Audience:

Development team and internal QA reviewers. This report **must not** be used as evidence of production readiness or as a basis for regulatory sign-off without further independent validation.

Report Generation:

Automatically generated on **17 April 2026 at 23:49:01** by the AeroFuel QA pipeline.
Python unittest framework with Flask Test Client, SQLite in-memory database.

Prepared By	Reviewed By	Approved By
AeroFuel QA Automation	_____	_____
System Generated	Date: _____	Date: _____